

CHRONIC & PERMANENT DISEASE MANAGEMENT

Diabetes | Hypertension | Heart Failure | Asthma | Epilepsy | Hypothyroidism

Clinical Guideline for AI Model Training Data — Version 1

Sources: WHO Primary Care Guidelines | MSF Clinical Guidelines | Hesperian Health Guides | WHO Integrated Management of Adult Illness

Chronic diseases require ongoing management even in conflict and low-resource settings. Treatment goals shift to symptom control, preventing life-threatening complications, and maintaining function when specialist care and medication supplies are limited or interrupted. This guideline focuses on non-pharmacological and low-resource approaches.

1. DIABETES MELLITUS

1.1 Recognition

Feature	Type 1 (Insulin-Dependent)	Type 2 (Non-Insulin-Dependent)
Onset	Usually sudden, younger patients	Gradual, usually adults over 40
Build	Often thin / normal weight	Often overweight
Key symptoms	Excessive thirst, frequent urination, rapid weight loss, blurred vision, fatigue	Same symptoms but milder; may be detected only at complication stage
Emergency sign	Kussmaul breathing, fruity breath, altered consciousness (DKA)	Confusion, extreme thirst, very high blood sugar (HHS)

1.2 Acute Complication: Hypoglycaemia (Low Blood Sugar)

Most dangerous acute complication in the field. Caused by missed meals, excess physical exertion, or over-treatment.

Stage	Signs & Symptoms	Low-Resource Management
Mild	Shaking, sweating, hunger, dizziness, palpitations	If conscious: give sugar in any form — 3–4 teaspoons of sugar dissolved in water, fruit juice, sweet tea, or honey. Repeat after 15 minutes if no improvement.
Moderate	Confusion, difficulty walking, aggressive behaviour	Same as mild if can swallow. Do NOT give food or drink if consciousness is impaired — risk of choking.
Severe	Unconscious, seizures, cannot swallow	Place in recovery position. Rub sugar, jam, or honey on inside of cheek and gums if semi-conscious. Urgent referral.

1.3 Acute Complication: Diabetic Ketoacidosis (DKA) — Type 1

- Signs: rapid deep breathing (Kussmaul), fruity or acetone breath, vomiting, abdominal pain, progressive drowsiness.
- Low-resource management: keep airway patent; position on side if drowsy; encourage oral fluids if conscious.
- Urgent referral — DKA is life-threatening and requires IV fluids and insulin.
- Monitor breathing and level of consciousness during transport.

1.4 Long-Term Low-Resource Care Principles

- Dietary control: reduce sugar, white flour, white rice; increase vegetables, legumes, whole grains.
- Regular physical activity: 30 minutes of walking most days if possible.
- Foot care: inspect feet daily; treat any wound promptly; never walk barefoot.

- Eye and kidney complications: monitor vision and urine output; refer any visual change or leg swelling.
- Fasting during conflict: advise patients not to skip meals; adjust food intake rather than eliminate.
- Infection risk: any infection worsens blood sugar control — treat infections early.

DO NOT stop insulin abruptly in Type 1 diabetes — life-threatening DKA will follow within hours.

DO NOT give anything by mouth to an unconscious diabetic patient.

DO NOT ignore foot wounds — infection can progress rapidly to gangrene.

2. HYPERTENSION (HIGH BLOOD PRESSURE)

2.1 Classification

Category	Systolic (mmHg)	Diastolic (mmHg)	Action
Normal	< 120	< 80	No action needed
Elevated	120–129	< 80	Lifestyle modification
Stage 1	130–139	80–89	Lifestyle + monitor
Stage 2	≥ 140	≥ 90	Treatment required
Hypertensive crisis	≥ 180	≥ 120	Emergency — immediate care

2.2 Hypertensive Emergency — Recognition

- Severe headache, blurred vision, chest pain, shortness of breath, confusion, facial droop, or one-sided weakness.
- Any blood pressure ≥ 180/120 mmHg with organ damage symptoms = hypertensive emergency.
- Stroke symptoms with high blood pressure: act as stroke emergency (see Neurological Emergencies guideline).

2.3 Non-Pharmacological Management (Low-Resource Setting)

- Reduce salt intake — avoid adding salt to food; avoid preserved or canned foods when possible.
- DASH-style diet: increase vegetables, fruit, and legumes; reduce animal fat.
- Weight reduction if overweight: even modest weight loss reduces blood pressure.
- Physical activity: 30 minutes of brisk walking most days reduces systolic BP by 5–8 mmHg.
- Reduce stress: relaxation techniques, breathing exercises, social support.
- Avoid alcohol and tobacco.
- Monitor blood pressure regularly if a device is available.
- Continued medication supply is critical — interruption causes dangerous rebound hypertension.

DO NOT abruptly stop antihypertensive medications — severe rebound hypertension can cause stroke.

Hypertensive emergency with neurological signs = treat as stroke until proven otherwise.

High blood pressure alone without symptoms does NOT require emergency treatment — reduce gradually.

3. HEART FAILURE

3.1 Recognition

Symptom	Description	Severity Indicator
Breathlessness	Shortness of breath on exertion; later at rest or lying flat	Lying flat impossible = severe
Leg oedema	Swelling of ankles and legs, pitting on pressure	Swelling above knee = severe
Fatigue	Severe tiredness with minimal effort	Unable to perform daily activities
Cough	Persistent cough, may produce frothy or pink-tinged sputum	Pink frothy sputum = acute pulmonary oedema

Rapid heartbeat	Palpitations, irregular pulse	Irregular pulse = refer urgently
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3.2 Acute Pulmonary Oedema — Emergency Management

STEP 1	POSITION
	Sit the patient upright — legs dangling down if possible. Never lay flat. Upright position reduces fluid in lungs.
STEP 2	OXYGEN
	Administer high-flow oxygen if available. Fan the patient's face if no oxygen.
STEP 3	REDUCE EXERTION
	Complete rest. No physical effort whatsoever until stabilised.
STEP 4	RESTRICT FLUIDS
	Do not give oral fluids beyond small sips. No IV fluids unless specifically required.
STEP 5	REFER URGENTLY
	Acute pulmonary oedema is life-threatening. Arrange emergency evacuation.

3.3 Ongoing Low-Resource Management

- Salt restriction: reduce sodium intake — most important non-pharmacological measure.
- Fluid restriction: in severe heart failure, limit fluid intake to 1–1.5 litres per day.
- Elevate legs: when resting, elevate legs above heart level to reduce oedema.
- Daily weight monitoring: sudden weight gain of > 1 kg overnight indicates fluid retention — restrict fluids and rest.
- Avoid exertion in heat: hot weather increases cardiovascular strain.
- Sleeping position: sleep with upper body elevated on folded clothing or blankets.
- Avoid lying flat: use inclined sleeping position if orthopnoea (breathlessness lying flat) is present.

DO NOT lay a patient with acute breathlessness flat — worsens pulmonary oedema immediately.

DO NOT give excess IV fluids to a patient with known heart failure.

4. EPILEPSY — SEIZURE MANAGEMENT

4.1 Types of Seizures

Type	Appearance	Duration
Generalised tonic-clonic (grand mal)	Stiffening of whole body followed by jerking of arms and legs; loss of consciousness; may bite tongue; urinary incontinence	Usually 1–3 minutes
Absence seizure (petit mal)	Sudden brief staring, unresponsive, eyelid fluttering; no fall	5–30 seconds
Focal (partial) seizure	Jerking limited to one limb or side; may spread; may or may not lose consciousness	Seconds to minutes
Status epilepticus	Continuous seizure lasting > 5 minutes OR two seizures without recovery between	Medical emergency

4.2 First Aid During a Seizure

STEP 1	PROTECT FROM INJURY
	Clear the area of hard or sharp objects. Place something soft under the head — folded clothing. Do NOT restrain the person.

STEP 2	TIME THE SEIZURE
	Note exact time seizure began. If > 5 minutes, this is status epilepticus — life-threatening emergency.
STEP 3	AIRWAY
	Do NOT put anything in the mouth. Do NOT try to hold the tongue. Tongue biting cannot be prevented by finger insertion and this causes injury to helper.
STEP 4	RECOVERY POSITION
	When jerking stops, place patient on their side (recovery position) to prevent aspiration of saliva or vomit.
STEP 5	POST-ICTAL PHASE
	Patient may be confused, sleepy, or agitated for minutes to hours after seizure. Reassure. Do not leave alone.

4.3 Status Epilepticus (Seizure > 5 Minutes) — Emergency

- Seizure lasting > 5 minutes or repeated seizures without recovery = status epilepticus.
- Maintain airway; recovery position; high-flow oxygen if available.
- Prevent injury; keep environment safe.
- Refer immediately — requires emergency medical intervention.
- If diazepam rectal solution is available per protocol: use per MSF guidelines. No pharmacological guidance included here.

DO NOT restrain seizure movements — causes fractures and dislocations.

DO NOT place objects or fingers in the mouth during a seizure.

DO NOT give food or water until patient is fully conscious.

Seizure > 5 minutes = life-threatening emergency — seek help immediately.

5. HYPOTHYROIDISM

5.1 Recognition

- Fatigue, weakness, feeling cold, constipation, dry skin, hair loss, weight gain, slow heartbeat.
- In severe or untreated cases: puffy face, slowed thinking, depression, hoarse voice.
- Myxoedema coma (extreme hypothyroidism): confusion, hypothermia, very slow heartbeat, decreased consciousness.
- In conflict settings: iodine deficiency is a major cause — goitre (enlarged thyroid) may be visible.

5.2 Low-Resource Management

- Iodine deficiency hypothyroidism: increase iodine intake — iodised salt, seafood, dairy when available.
- Keep patient warm: hypothyroid patients cannot regulate body temperature — blankets, avoid cold environments.
- Myxoedema coma: medical emergency. Keep airway patent, warm patient, urgent referral.
- Thyroid hormone replacement (levothyroxine) is the definitive treatment — requires medical prescription and supply.
- Do not abruptly stop thyroid medication if supply is interrupted — taper gradually and seek resupply.

6. RED FLAGS — CRITERIA FOR URGENT REFERRAL

Red Flag Criterion	Clinical Significance / Action
DKA / HHS	Rapid breathing, fruity breath, vomiting, drowsiness — refer immediately
Severe hypoglycaemia	Unconscious diabetic patient — refer after first aid

Hypertensive emergency	BP $\geq$ 180/120 with headache, visual change, chest pain, or neurological signs
Acute pulmonary oedema	Severe breathlessness, pink frothy sputum, unable to lie flat
Status epilepticus	Seizure lasting > 5 minutes — life-threatening
Myxoedema coma	Hypothermia + confusion + bradycardia in known hypothyroid patient
Chest pain + breathlessness	Possible myocardial infarction — refer urgently regardless of resources

REF. REFERENCES

- [1] WHO. Primary Care Guidelines for Low-Resource Settings. Geneva: WHO.
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- [5] WHO. mhGAP Intervention Guide for Mental, Neurological and Substance Use Disorders. Version 2. Geneva: WHO, 2016.
- [6] Epilepsy Foundation. First Aid for Seizures. <https://www.epilepsy.com>

This document was compiled strictly from the sources listed above for use as AI model training data in clinical emergency assessment. No clinical decisions should be made based on this document alone without consulting qualified medical professionals.